MISUN KIM

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RESEARCH INTERESTS

I study why alignment interventions and safety behaviour that work in English fail in other languages, and where in the model the failure arises. My work separates the intervention from the input it acts on, and tests whether automated safety evaluation is equally reliable across languages.

EDUCATION

MSc Computer Science with Speech and Natural Language Processing
University of Sheffield | Distinction (73.17/100) | Dissertation: 78/100 **Sep 2025 - Sep 2026**

- Dissertation: Cross-Lingual Activation Steering in Large Language Models: Representational Alignment and Behavioural Transfer

BEng Industrial and Management Engineering; BA Central Asian Studies
Hankuk University of Foreign Studies | South Korea | GPA 3.94/4.50 **Mar 2019 - Feb 2024**

- Kazakh language and regional studies · First Distinction Scholarship (2023)

RESEARCH EXPERIENCE

MSc Dissertation Research — Cross-Lingual Transfer of Activation Steering
University of Sheffield | Supervisor: Dr Cass Zhao **Feb 2026 - Sep 2026**

- Investigated the robustness of activation-level interventions in multilingual LLMs using Contrastive Activation Addition (CAA), testing whether English-derived steering vectors transfer consistently across three model families and five languages.
- Identified a representation–behaviour gap: high steering-vector similarity did not reliably predict the magnitude or direction of behavioural transfer across models and tasks, concluding that high representational similarity is not sufficient to guarantee comparable behavioural effects, and leaving the question of necessity unresolved.
- Designed layer-wise evaluations with bootstrap confidence intervals, held-out sets, and Gaussian and permutation random-vector baselines to characterise failure modes and distinguish systematic effects from noise.
- Conducted a separate cross-task evaluation using English-derived truthfulness vectors on TruthfulQA, finding a descriptive cross-task contrast across models and behaviours: Qwen3.5-9B transferred sycophancy but not truthfulness, OLMo3-7B-Instruct the reverse.
- Follow-up study (post-MSc): built a full 5×5 source–target transfer matrix with a norm-matched control, separating how well each language supplies a steering vector from how well its inputs respond.

Undergraduate Researcher **Mar 2023 - Nov 2023**
ICT Mentoring Project, Ministry of Oceans and Fisheries | Supervisor: Gyu-Young Lee (KAIST)

- First-authored LSTM study on container freight forecasting across five universities, reducing RMSE by 71% over the baseline (published ACK 2023)

PUBLICATIONS & MANUSCRIPTS

- **Kim, M.**, Kim, Y., Kim, E., Lee, B., Han, Y., & Lee, G. (2023). Implementation of Container Volume Prediction Technology using Deep Learning. ACK Conference Proceedings, 30(2).
- Manuscript in preparation (2027). Cross-lingual and cross-task generalisation of representation-level interventions in multilingual LLMs.

KEY PROJECTS

Post-Training Safety Evaluation Framework **Aug 2026 - Present**

- Built a reproducible framework to investigate cross-lingual transfer of safety refusal behaviour across OLMo 3 post-training checkpoints.
- Evaluated 200 matched EN/KO prompts per language, finding ~80% vs. ~10% scorer-measured harmful-request refusal rates, with a large gap persisting under human validation.
- Conducted stratified human validation (70 responses) with bootstrap and Bayesian uncertainty analysis, revealing opposing patterns of safety-judge errors across languages.

ESG Compliance Checker **Mar 2026 - Aug 2026**

- Built an end-to-end compliance verification system over 11 corporate ESG reports, mapping regulatory criteria to retrieved evidence and producing traceable covered/partial/missing judgements using hybrid retrieval and LLM-assisted verification.
- Developed and audited a manually annotated retrieval gold set; hybrid retrieval with reranking improved Hit@5 from 83.3% to 100% and MRR from 0.628 to 0.819 over BM25-only retrieval.
- Used retrieval and verifier error analysis to identify annotation gaps and evidence-selection failures, then modularised the system with FastAPI, automated testing, Docker, and a separate Streamlit interface.

SKILLS

Programming: Python, SQL
ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn
LLM Alignment & Evaluation: Activation steering (CAA), representation analysis, cross-lingual and cross-task evaluation, post-training safety evaluation, LLM-as-judge validation, human annotation and error analysis
Experimental Methods: Controlled experiments, bootstrap confidence intervals, Bayesian uncertainty analysis, random-vector baselines
Research Infrastructure: HPC/SLURM, Weights & Biases, Git, GitHub Actions (CI), Docker, FastAPI, Flask, ChromaDB

WORK EXPERIENCE

Sustainability Research Analyst **Mar 2025 - May 2025**
Eon Consulting Group | South Korea

- Contributed to 170+ page sustainability reports for major public and financial organisations, conducting TNFD climate scenario analysis and developing a CSDDD-aligned supply-chain risk assessment framework.

Field Service Manager (Data & Operations)
Payhere | South Korea **Mar 2024 - Aug 2024**

- Built workflow automation using the Slack API, Google Sheets, and Looker Studio to analyse 10,000+ service records, reducing unnecessary field visits by 17%.

TEACHING EXPERIENCE

Language Tutor (Private) **Mar 2021 - Nov 2025**

- Provided individualized Korean/English instruction to 5+ learners aged 6–18, from early literacy to CSAT-level advanced reading

LANGUAGES

Korean (Native), English (Fluent), Spanish (Intermediate), Kazakh (Intermediate)